WI Generator: Model Summary, all PCPs

5 parts: 289661208202 (1 stage), 290058600101 (4 stages), 265472208237 (1 stage), 511830100101 (1 stage), 544283753302 (1 stage) · 5 models · LLM judge (`wi_judge_single_prompt.md`) · Compiled 11 November 2026

5	8	5	40	0	$0.185
parts (PCPs)	stage-WIs per model	models	WIs judged in total	WIs that pass the judge	total generation cost

Summary

- 40 WIs were judged: 5 models × 8 stage-WIs from 5 parts (one part with 4 stages, four with 1). **None passes the judge.** Every WI fails at least one gate: a contradicted critical fact, a critical hallucination, or critical recall below 0.95.
- **Luna-5.6 is the strongest model.** It has a mean overall score of 0.900, is best on 7 of 8 WIs and 4 of 5 parts, and is the only model whose pooled critical recall clears the 0.95 gate (0.952). **Luna-6 is a close second** (0.867). It wins on 289661208202 and costs less than half as much ($0.0041 vs $0.0085 per WI).
- The other three models are far behind. mistral-small averages 0.566, but its scores range from 0.27 to 0.93 because it often stops after one operation. gpt-4o-mini (0.401) and Nano-5 (0.256) write English on every part and copy the generator prompt's welding example instead of reading the PPAP.
- Almost none of the Luna models' failures are in the process content. Every critical fact they contradict is a header field (Stage No., revision or date). Of the critical additions that were checked against the PPAP, 193 of 195 are real PPAP data that the golden leaves out. If PPAP-backed additions were exempt, Luna-5.6 would pass 2 of its 7 tagged WIs, and clear header rules would remove most of the remaining failures.
- The prompt, the judge and the golden WIs need fixing more than the model choice does. Remove the filled-in example from `prompt_builder.py` , write explicit header rules, let the judge accept additions that come from the PPAP, and review the golden WIs with their author. Then re-run all five parts.

Recommendation

Use **Luna-5.6** as the lead model. It is ahead on 7 of 8 WIs and is the only model with pooled critical recall above the gate. At $0.0085 per WI, it costs about $0.44 more per 100 WIs than Luna-6, which is small next to the cost of an operator following a wrong WI. Keep **Luna-6** as the cost alternative: its process content is as good, and most of its gap comes from the Stage No. header error, which a prompt rule should fix. Drop gpt-4o-mini, Nano-5 and mistral-small. Make the final decision only after re-running both Luna models with the fixed prompt and judge.

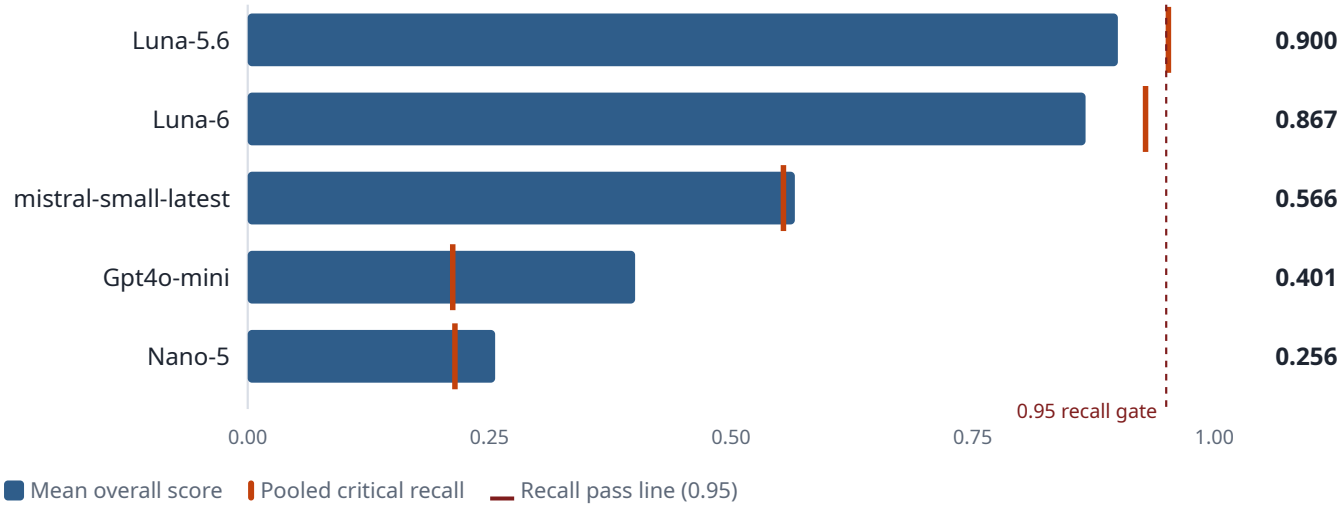
Leaderboard

#	Model	Mean overall per part	Range over 8 WIs	Critical recall pooled	Crit. contra-dicted	Crit. halluc-inations (in PPAP)*	Facts matched	Best on parts / WIs	Hindi	Cost / WI
1	**Luna-5.6** openai / gpt-5.6-luna	**0.900**	0.85–0.96	0.952	11	116 (97/99)	750/827	4 / 7	5/5	$0.0085
2	**Luna-6** openai / gpt-6-luna	**0.867**	0.84–0.92	0.929	10	109 (96/96)	720/827	1 / 1	5/5	$0.0041

3	**mistral-small-latest** mistral / mistral-small-latest	**0.566**	0.27–0.93	0.554	16	63 (43/55)	483/827	0 / 0	2/5	$0.0049
4	**Gpt4o-mini** openai / gpt-4o-mini	**0.401**	0.31–0.54	0.212	24	26 (8/21)	288/827	0 / 0	0/5	$0.0039
5	**Nano-5** openai / gpt-5-nano	**0.256**	0.17–0.47	0.214	28	81 (41/70)	256/827	0 / 0	0/5	$0.0016

Mean overall is the average of the five per-part scores (each part counts once; 290058600101 uses the mean of its four stages, as in its own report). Critical recall is pooled over every critical golden fact in all 8 stage-WIs. Hindi = parts where the output language was compliant. Cost / WI = total generation cost / WIs generated. * Bracket: critical hallucinations whose content the judge found in the PPAP, out of those tagged. 289661208202 was judged before tagging was added, so its 54 additions are not counted in the bracket.

Mean overall score vs. pooled critical recall



Results by part

Part	Description	Stages	Golden facts	Best model	Best overall	Per-part report
289661208202	CROSSMEMBER LH 2DR XENON CAB	1	109	Luna-6	0.873	MODEL_COMPARISON_289661208202.pdf
290058600101	BKT ASSY JACK MTG	4	377	Luna-5.6	0.930	MODEL_COMPARISON_290058600101.pdf
265472208237	SUPPORT LH/RH DOOR OUTER HANDLE	1	98	Luna-5.6	0.899	MODEL_COMPARISON_265472208237.pdf
511830100101	BRACKET ASSY,ACCELERATOR PEDAL MTG	1	108	Luna-5.6	0.915	MODEL_COMPARISON_511830100101.pdf
544283753302	BRACKET SUCTION PIPE	1	135	Luna-5.6	0.904	MODEL_COMPARISON_544283753302.pdf

Overall score per stage-WI

Cell colours: ≥0.90 0.60–0.89 0.30–0.59 <0.30 Outlined cell = best model on that WI

Model	289661208202	290058600101				265472208237	511830100101	544283753302	Mean per part
	S01	S01	S02	S03	S04	S01	S01	S01	
Luna-5.6	0.853	0.959	0.930	0.928	0.902	0.899	0.915	0.904	**0.900**
Luna-6	0.873	0.915	0.879	0.883	0.856	0.847	0.892	0.837	**0.867**
mistral-small-latest	0.444	0.626	0.928	0.810	0.297	0.708	0.743	0.269	**0.566**
Gpt4o-mini	0.432	0.419	0.312	0.500	0.538	0.377	0.392	0.361	**0.401**
Nano-5	0.174	0.426	0.227	0.466	0.167	0.178	0.410	0.195	**0.256**

Critical recall per part

Green = clears the 0.95 recall gate.

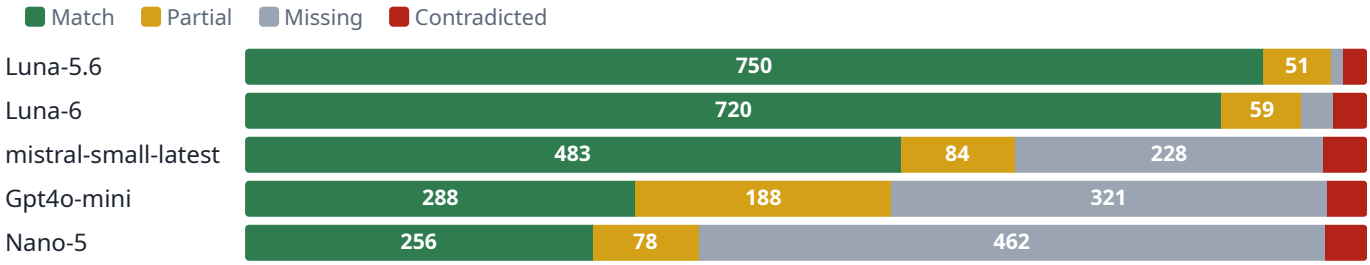
Model	289661208202	290058600101	265472208237	511830100101	544283753302	Pooled
Luna-5.6	0.931	0.948	0.946	0.968	0.974	**0.952**
Luna-6	0.917	0.943	0.875	0.968	0.910	**0.929**
mistral-small-latest	0.389	0.711	0.571	0.742	0.154	**0.554**
Gpt4o-mini	0.167	0.278	0.214	0.129	0.154	**0.212**
Nano-5	0.125	0.299	0.125	0.290	0.090	**0.214**

Where the models differ

Section scores, pooled over all 827 golden facts

Section (weight)	Luna-5.6	Luna-6	mistral-small-latest	Gpt4o-mini	Nano-5
Steps (0.35)	0.98	0.98	0.51	0.27	0.15
Finished QC (0.15)	0.99	0.98	0.60	0.46	0.25
Rework (0.15)	0.65	0.43	0.35	0.21	0.26
Incoming QC (0.10)	0.89	0.90	0.80	0.74	0.18
Header (0.10)	0.90	0.84	0.78	0.60	0.56
Tooling (0.10)	0.98	0.96	0.61	0.32	0.20
Legend (0.05)	1.00	1.00	1.00	1.00	1.00
Mean overall	**0.900**	**0.867**	**0.566**	**0.401**	**0.256**

Verdicts on all 827 golden facts



Why nothing passes

Model	WIs passed	WIs with a contradicted critical fact	WIs with a critical hallucination	WIs with critical recall < 0.95	Would pass if PPAP-backed additions were exempt**
Luna-5.6	0/8	5	8	4	**2**/7
Luna-6	0/8	8	8	5	**0**/7
mistral-small-latest	0/8	8	8	7	**0**/7
Gpt4o-mini	0/8	8	8	8	**0**/7
Nano-5	0/8	8	8	8	**0**/7

A WI can fail more than one gate. ** Re-applies the pass rule with critical additions that the judge found in the PPAP no longer counted; only possible on the 7 WIs whose additions were tagged (not 289661208202).

- **Contradicted critical facts.** All 21 contradictions from the two Luna models are header fields. Luna-6 writes Stage No. as an operation-code range (e.g. “PP-010 to PP-130”) on all 8 WIs, so every Luna-6 WI fails this gate. Luna-5.6 takes the PCP form's revision and date instead of the part's on 289661208202 and 290058600101 (stages 03 and 04). Some golden Stage No. values (“Multiple”, “1-5”, “Multiple Stages”) follow no rule that any model could derive.
- **Critical hallucinations.** Every model fails this gate on all 8 WIs. For the Luna models, 193 of 195 tagged additions are in the PPAP: minimum shut heights, strokes, pressures, extra CD dimensions, form references (F-PRD-17, F-QMS-6) and sampling frequencies. Only two are not supported by the PPAP, both from Luna-5.6: “Blanking Tool” on 290058600101 S03, and radii R3 / R4.5 / R5 on 265472208237. For the weak models, many additions are invented or copied from the prompt (29 of Nano-5's 70 tagged additions are not in the PPAP).
- **Critical recall.** Luna-5.6 falls below 0.95 on 4 WIs and Luna-6 on 5. The facts they miss are mostly golden values that the PPAP does not contain: the ±0.05 mm and ±0.001 mm thickness tolerances, “UNIT-3”, and the header fields above.

Cost and quality



Model	Cost / WI	Cost / 100 WIs	Total for this eval	LLM time / WI	Output tokens / WI	max_tokens used	Score per cent
Luna-5.6	$0.0085	$0.85	$0.0681	29.9 s	3,126	4,000 / 16,000	1.06

Luna-6	$0.0041	$0.41	$0.0329	33.9 s	3,458	4,000 / 16,000	2.11
mistral-small-latest	$0.0049	$0.49	$0.0396	11.6 s	1,370	16,000	1.14
Gpt4o-mini	$0.0039	$0.39	$0.0315	6.7 s	605	4,000 / 16,000	1.02
Nano-5	$0.0016	$0.16	$0.0131	10.1 s	1,114	4,000	1.56

Score per cent = mean overall score / cost per WI in US cents. Costs come from each run's own cost report.

gpt-4o-mini costs about the same as Luna-6 but scores 0.47 lower. Nano-5 is the cheapest, but its output cannot be used. At 100 WIs, the price difference between the two Luna models is $0.44 ($0.85 vs $0.41). Luna-5.6 ran with `max_tokens` 4,000 on four of the five parts and used up to 3,688 output tokens, so it was close to being cut off. Raise the limit to 16,000 for both Luna models.

Model profiles across all parts

#1 Luna-5.6 Lead candidate

Mean **0.900** Recall **0.952** Best on **7/8** WIs $0.0085/WI

STRENGTHS Best on 7 of 8 WIs and 4 of 5 parts, and the only model with pooled critical recall above 0.95 (0.952). It writes the whole process on every part, with presses, shut heights, strokes, dimensions, weld settings, torques and coating values bound to the right operation (steps 0.976, tooling 0.985, finished QC 0.989). It also has the best rework score of any model (0.645) and writes Hindi on all five parts.

WEAKNESSES

- 11 contradicted critical facts, all header fields. It takes the PCP form's revision and date (00, 11/11/17) instead of the part's on 289661208202 and 290058600101, and Stage No. is wrong on three WIs.
- 116 critical additions. 97 of the 99 tagged ones are in the PPAP; the two that are not are “Blanking Tool” (290058600101 S03) and radii R3 / R4.5 / R5 (265472208237).
- Costs 2.1× as much as Luna-6 ($0.0085 per WI). It ran with a 4,000-token limit on four parts and used up to 3,688 tokens.

#2 Luna-6 Keep testing

Mean **0.867** Recall **0.929** Best on **1/8** WIs $0.0041/WI

STRENGTHS A close second (0.867), and best on 289661208202 (0.873). Its process sections are as good as Luna-5.6's (steps 0.978, tooling 0.965, finished QC 0.982). All 96 of its tagged critical additions are in the PPAP, it writes Hindi on all five parts, and it costs about the same as gpt-4o-mini ($0.0041 per WI).

WEAKNESSES

- Writes Stage No. as an operation-code range (“PP-010 to PP-130”) on all 8 WIs, so every WI fails the contradiction gate. This is the main reason for its lower header score (0.84).
- Weaker rework (0.427). It rejects the part or gives vaguer actions where the golden repairs it.
- Critical recall is 0.929 pooled and below 0.95 on 5 of 8 WIs.

#3 mistral-small-latest Drop

Mean **0.566** Recall **0.554** Best on **0/8** WIs $0.0049/WI

STRENGTHS When it covers the whole stage, the result is usable: 0.928 on 290058600101 S02 and 0.743 on 511830100101, with correct press settings.

WEAKNESSES

- Its coverage is unreliable (scores range from 0.269 to 0.928). It wrote only the draw operation on 289661208202, stopped at raw-material receipt on 544283753302, and scored 0.297 on 290058600101 S04.
- Its language is not compliant on 3 of 5 parts. It invents tolerances and copies “UNIT-3” from the prompt example. 12 of its 55 tagged additions are not in the PPAP.
- Its pooled critical recall is 0.554.

#4 Gpt4o-mini Drop

Mean **0.401** Recall **0.212** Best on **0/8** WIs $0.0039/WI

STRENGTHS The header identifiers and incoming-material checks are usually correct.

WEAKNESSES

- Writes English on all five parts, although the run language was Hindi.
- Copies the prompt's welding steps (“Collect all loose parts”, “Clamp the fixture”) and gives an outline without values. It averages 605 output tokens per WI, against more than 3,000 for the Luna models.
- Pooled critical recall is 0.212, with 24 contradicted critical facts.

#5 Nano-5 Drop

Mean **0.256** Recall **0.214** Best on **0/8** WIs $0.0016/WI

STRENGTHS The cheapest model ($0.0016 per WI). The header and part-list identifiers are often correct.

WEAKNESSES

- Reproduces the prompt's spot-welding example almost word for word (welding fixture, 12 spot welds, “Gun No. 90”) instead of the press process.
- Writes English on all five parts, has 28 contradicted critical facts, and 29 of its 70 tagged additions are not in the PPAP.
- Lowest score on every part except 511830100101 (mean 0.256).

Recurring root causes

1. The prompt's filled-in welding example leaks into the output (all 5 parts)

The JSON example in `prompt_builder.py` is a complete spot-welding WI for a different part. gpt-4o-mini and Nano-5 copy it on every part. mistral-small copies its “UNIT-3” line area, and Luna-5.6 took its 11/11/17 date on 289661208202. Some golden WIs contain the same example values (“UNIT-3”, plastic mallet, rust remover), so copying the example is rewarded.

2. Header fields have no explicit rules

The PPAP has up to three candidate revisions and dates: the part's, the control plan's and the PCP form's. The prompt does not say which one to use, or what Stage No. should hold when one WI covers several operations. As a result, all 21 critical contradictions from the Luna models are header fields, and the golden WIs are inconsistent here too.

3. The judge counts PPAP-supported detail as hallucination

The judge compares only against the golden WI. Control-plan rows that the golden leaves out are counted as critical hallucinations even when they are in the PPAP, and this is 193 of the 195 tagged additions from the Luna

models. Because of this, the hallucination gate cannot tell the best models from the worst: every model fails it on every WI.

4. Some golden values cannot be derived from the PPAP

Some golden values do not appear in the source PPAP: line area “UNIT-3” (the PPAP says UNIT-II), the Stage No. values “Multiple”, “1-5” and “Multiple Stages”, the ±0.05 mm and ±0.001 mm tolerances, and the plastic-mallet, rust-remover and rust-preventive rework rows. Rework is the weakest section for every model; the best score is 0.645.

5. The weaker models do not extract the process

gpt-4o-mini writes about 600 output tokens per WI and Nano-5 about 1,100, against 3,100 to 3,500 for the Luna models. The token limit does not explain this, because they stop well before it. mistral-small often covers only the first operation of the stage. These models summarise or copy the prompt instead of turning the control plan into steps.

Recommendations

1. **Fix the prompt example.** Replace the filled-in welding WI in `prompt_builder.py` with placeholders or a neutral press-shop example, and tell the model never to copy example values.
2. **Add header rules to the prompt.** Stage No. is the WI's stage number, never an operation-code range. Revision and date come from the control plan's Rev.No. / Date (Rev.), not from the form. Line area is the unit given in the PPAP. An LH/RH pair gets one part-list row per part number.
3. **Fix the judge.** Add a source-supported class (e.g. `SOURCE_SUPPORTED_ADDITION`) for additions found in the PPAP, and keep the hallucination gate for content that is in neither the golden nor the PPAP. Re-tag 289661208202, which was judged before PPAP tagging was added.
4. **Review the golden WIs with their author:** UNIT-3, the Stage No. values, the tolerances, the rework rows, and the control-plan rows that the golden leaves out. Then update `golden_facts_Stage_NN.json` and re-score.
5. **Model choice:** make Luna-5.6 the lead model and keep Luna-6 as the lower-cost alternative. Drop gpt-4o-mini, Nano-5 and mistral-small. Set `max_tokens` to 16,000 for both Luna models.
6. **Re-run all five parts** with the fixed prompt and judge, using at least 3 generations per model, before making the final choice. The current gap between the two Luna models (0.033) is within the variation between parts.

Method and sources

Each part was evaluated separately with the runbook in `WI_EVAL_RUNBOOK.md`: every golden WI was split into atomic facts, Claude applied `wi_judge_single_prompt.md` to each generated WI, and `eval_tools/score.py` computed the scores from the verdicts. This summary re-reads the same `judge_result_*.json` files with `eval_tools/build_summary.py`; no verdict was changed. Facts are weighted 3 / 2 / 1 (critical / major / minor) and sections Steps 0.35, Finished QC 0.15, Rework 0.15, Incoming QC 0.10, Header 0.10, Tooling 0.10, Legend 0.05. A WI passes only with 0 contradicted critical facts, 0 critical hallucinations and critical recall of at least 0.95.

Limits. Five parts, 8 stage-WIs and one generation per model, scored by one judge pass. Generation settings were not uniform: max_tokens was 4,000 on some runs and 16,000 on others. The golden WIs have known problems (see root cause 4) that lower every model's score. Treat the gap between the two Luna models as indicative. The gap between the Luna models and the other three is large and consistent on every part. An earlier embedding-based comparison (MODEL_SUMMARY_ALL_PCPs.docx, 30 Oct) gave the same top three in the same order: Luna-5.6, Luna-6, mistral-small. Full findings, evidence and judgement calls are in each part's own report (listed under Results by part).